

Photo Credits for Volume 2

Except as specifically noted below or in a parenthetical credit in the caption of a figure, all the illustrations in this book are under my own copyright, and are copyleft licensed under the same license as the rest of the book.

In some cases it's clear from the date that the figure is public domain, but I don't know the name of the artist or photographer; I would be grateful to anyone who could help me to give proper credit. I have assumed that images that come from U.S. government web pages are copyright-free, since products of federal agencies fall into the public domain. I've included some public-domain paintings; photographic reproductions of them are not copyrightable in the U.S. (*Bridgeman Art Library, Ltd. v. Corel Corp.*, 36 F. Supp. 2d 191, S.D.N.Y. 1999).

When "PSSC Physics" is given as a credit, it indicates that the figure is from the first edition of the textbook entitled *Physics*, by the Physical Science Study Committee. The early editions of these books never had their copyrights renewed, and are now therefore in the public domain. There is also a blanket permission given in the later PSSC College Physics edition, which states on the copyright page that "The materials taken from the original and second editions and the Advanced Topics of PSSC PHYSICS included in this text will be available to all publishers for use in English after December 31, 1970, and in translations after December 31, 1975."

Credits to Millikan and Gale refer to the textbooks *Practical Physics* (1920) and *Elements of Physics* (1927). Both are public domain. (The 1927 version did not have its copyright renewed.) Since it is possible that some of the illustrations in the 1927 version had their copyrights renewed and are still under copyright, I have only used them when it was clear that they were originally taken from public domain sources.

In a few cases, I have made use of images under the fair use doctrine. However, I am not a lawyer, and the laws on fair use are vague, so you should not assume that it's legal for you to use these images. In particular, fair use law may give you less leeway than it gives me, because I'm using the images for educational purposes, and giving the book away for free. Likewise, if the photo credit says "courtesy of ...," that means the copyright owner gave me permission to use it, but that doesn't mean you have permission to use it.

Photo credits to NEI refer to photos from the National Eye Institute, part of the National Institutes of Health, <http://www.nei.nih.gov/photo/>. "Items here are not copyrighted. However, we do ask that you credit as follows: National Eye Institute, National Institutes of Health (except where indicated otherwise)."

Cover Spider: Wikimedia Commons user Opoterser, CC-BY-SA. **Contents** See photo credits below, referring to the places where the images appear in the main text.

?? Faraday: Painting by Thomas Phillips, 1842. **?? Binoculars:** Wikimedia commons, CC-BY-SA. **?? Porro prisms:** Redrawn from a figure by Wikipedia user DrBob, CC-BY-SA. **Contents** **Insect's eye:** Wikimedia Commons, CC-BY-SA, user Reytan. **Contents** **Man's eye:** NEI. **537 Salamander:** Redrawn from an 18th century illustration by J.D. Meyer. **537 Hand:** Redrawn from a photo by Wikimedia Commons user Evan-Amos, CC-BY-SA. **575 Lightning:** C. Clark/NOAA photo library, public domain. **581 Knifefish:** Courtesy of Greg DeGreef. **582 Ampère:** Millikan and Gale, 1920. **585 Volta:** Millikan and Gale, 1920. **590 Ohm:** Millikan and Gale, 1920. **592 MRI:** Wikimedia Commons user Tomas Vendis, GFDL license. **595 Resistors:** Wikipedia user Afrank99, CC-BY-SA. **608 Crab:** From a photo by Hans Hillewaert, CC-BY-SA. **619 LP record:** Felipe Micaroni Lalli, CC-BY-SA. **619 Printed circuit board:** Bill Bertram, Wikipedia user Pixel8, CC-BY licensed. **634 GPS:** Wikipedia user HawaiianMama, CC-BY license. **634 Atomic clock on plane:** Copyright 1971, Associated press, used under U.S. fair use exception to copyright law. **635 Football pass:** Wikipedia user RMelon, CC-BY-SA licensed. **640 LIGO:** Wikipedia user Umptanum: "This image is copyrighted. The copyright holder has irrevocably released all rights to it." **647 Hammerhead shark:** Wikimedia Commons user Littlegreenman, public domain. **653 Topographical maps:** Flat maps by USGS, public domain; perspective map by Wikipedia user Kbh3rd, CC-BY-SA licensed. **656 Avalanche transceiver:** Human figure: based on Pioneer 10 plaque, NASA, public domain. Dipole field: Wikimedia Commons user Geek3, CC-BY. **669 Cow:** Redrawn from a photo by Wikimedia Commons user Cgoodwin, CC-BY-SA. **669 Car:** Redrawn from a photo by Wikimedia Commons user Auregann, CC-BY-SA. **670 Hell:** Hieronymus Bosch, public domain. **671 Surveyors:** Don Swanson, public domain; Owais Khursheed, CC-BY-SA; Paul Peterson, public domain; British Royal Navy, public domain. **671 Pigeon:** Wikimedia commons user nitramtrebla, CC-BY-SA. **671 Mushroom cloud:** USGS, public domain. **673 S. Iijima and K. Fujiwara, "An experiment for the potential blue shift at the Norikura Corona Station," Annals of the Tokyo Astronomical Observatory, Second Series, Vol. XVII, 2 (1978) 68:** Used here under the U.S. fair use doctrine. **673 Horse:** From a public-domain photo by Eadweard Muybridge, 1872. **673 Satellite:** From a public-domain artist's conception of a GPS satellite, product of NASA. **674 Joan of Arc holding banner:** Ingres, 1854. **674 Joan of Arc interro-**

gated: Delaroché, 1856. **684** *Muon storage ring at CERN*: (c) 1974 by CERN; used here under the U.S. fair use doctrine. **686** *Colliding nuclei*: courtesy of RHIC. **708** *Galaxies*: Ville Koistinen, CC-BY-SA. **708** *Machine gunner's body*: Redrawn from a public-domain photo by Cpl. Sheila Brooks. **708** *Machine gunner's head*: Redrawn from a sketch by Wenceslas Hollar, 17th century. **712** *Hertz*: Public domain contemporary photograph. **715** *Swan Lake*: Peter Gerstbach, GFDL 1.2. **716** *C.S. Wu*: Smithsonian Institution, believed to be public domain. **736** *Capacitors*: Wikipedia user de:Benutzer:Honina, CC-BY-SA licensed. **736** *Inductors*: Wikipedia user de:Benutzer:Honina, CC-BY-SA licensed. **753** *Curies*: Harper's Monthly, 1904. **757** *Robert Millikan*: Clark Millikan, 1891, public domain. **760** *J.J. Thomson*: Millikan and Gale, 1920.. **765** *Becquerel*: Millikan and Gale, 1920. **765** *Becquerel's photographic plate*: public domain. **767** *Nuclear fuel pellets*: US DOE, public domain. **768** *Rutherford*: public domain. **780** *Nuclear power plant*: Wikipedia user Stefan Kuhn, CC-BY-SA licensed. **785** *GAMMASPHERE*: courtesy of C.J. Lister/R.V.F. Janssens. **785** *H bomb test*: public domain product of US DOE, Ivy Mike test. **785** *Fatu Hiva Rainforest*: Wikipedia user Makemake, CC-BY-SA licensed. **785** *fusion reactor*: "These images may be used free of charge for educational purposes but please use the acknowledgement 'photograph courtesy of EFDA-JET'". **785** *Sun*: SOHO (ESA & NASA). **788** *Chernobyl map*: CIA Handbook of International Economic Statistics, 1996, public domain. **789** *Fifty-foot woman*: Public domain due to nonrenewal of copyright. **789** *Zombies*: Public domain due to an error by the distributor in failing to place a copyright notice on the film. **792** *Horses*: (c) 2004 Elena Filatova. **792** *Polar bear*: U.S. Fish and Wildlife Service, public domain. **793** *Crab Nebula*: "ESO Press Photos may be reproduced, if credit is given to the European Southern Observatory.". **799** *Eclipse*: 1919, public domain. **799** *Newspaper headline*: 1919, public domain. **801** *Photo of PET scanner*: Wikipedia user Hg6996, public domain. **801** *Ring of detectors in PET scanner*: Wikipedia user Damato, public domain. **801** *PET body scan*: Jens Langner, public domain. **820** *Einstein's ring*: I have lost the information about the source of the bitmapped image. I would be grateful to anyone who could put me in touch with the copyright owners.. **823** *Artificial horizon*: NASA, public domain. **824** *Pound and Rebka photo*: I presume this photo to be in the public domain, since it is unlikely to have had its copyright renewed. **825** *Earth*: NASA, Apollo 17. Public domain. **825** *Orion*: Wikipedia user Mouser, GFDL. **825** *M100*: European Southern Observatory, CC-BY-SA. **825** *Supercluster*: Wikipedia user Azcolvin429, CC-BY-SA. **828** *Globular cluster*: Hubble Space Telescope. Hubble material is copyright-free and may be freely used as in the public domain without fee, on the condition that NASA and ESA is credited as the source of the material. The material was created for NASA by STScI under Contract NAS5-26555 and for ESA by the Hubble European Space Agency Information Centre. **831** *Galaxies*: Hubble Space Telescope. Hubble material is copyright-free and may be freely used as in the public domain without fee, on the condition that NASA and ESA is credited as the source of the material. The material was created for NASA by STScI under Contract NAS5-26555 and for ESA by the Hubble European Space Agency Information Centre. **831** *Cosmic microwave background image*: NASA/WMAP Science Team, public domain. **837** *Rays of sunlight*: Wikipedia user PiccoloNamek, CC-BY-SA. **840** *Jupiter and Io*: NASA/JPL/University of Arizona. **849** *Ray-traced image*: Gilles Tran, Wikimedia Commons, public domain. **855** *Narcissus*: Caravaggio, ca. 1598. **857** *Praizinoscope*: Thomas B. Greenslade, Jr.. **859** *The Sleeping Gypsy*: H. Rousseau, 1897. **862** *Flower*: Based on a photo by Wikimedia Commons user Fir0002, CC-BY-SA. **862** *Moon*: Wikimedia commons image. **863** *Ladybug*: Redrawn from a photo by Wikimedia Commons user Gilles San Martin, CC-BY-SA. **883** *Fish-eye lens*: Martin Dürschnabel, CC-BY-SA. **884** *Hubble space telescope*: NASA, public domain. **888** *Anamorphic image*: Wikipedia user Istvan Orosz, CC-BY. **891** *Flatworm*: CC-BY-SA, Alejandro Sánchez Alvarado, Planaria.neuro.utah.edu. **891** *Nautilus*: CC-BY-SA, Wikimedia Commons user Opencage, opencage.info. **891** *Human eye*: Joao Estevao A. de Freitas, "There are no usage restrictions for this photo". **892** *Cross-section of eye*: NEI. **892** *Eye's anatomy*: After a public-domain drawing from NEI. **896** *Water wave refracting*: Original photo from PSSC. **898** *Ulcer*: Wikipedia user Aspersions, CC-BY-SA. **902** *Jumping spider*: Photo: Wikipedia user Opoterser, CC-BY-SA. Line art: redrawn from M.F. Land, J. Exp. Biol. 51 (1969) 443. **913** *Pleiades*: NASA/ESA/AURA/Caltech, public domain. **914** *Diffraction of water waves*: Assembled from photos in PSSC. **914** *Counterfactual lack of diffraction of water waves*: Assembled from photos in PSSC. **914** *Diffraction of water waves*: Assembled from photos in PSSC. **915** *Scaling of diffraction*: Assembled from photos in PSSC. **916** *Huygens*: Contemporary painting?. **917** *Diffraction of water waves*: Assembled from photos in PSSC. **918** *Young*: Wikimedia Commons, "After a portrait by Sir Thomas Lawrence, From: Arthur Shuster & Arthur E. Shipley: Britain's Heritage of Science. London, 1917". **923** *Single-slit diffraction of water waves*: PSSC. **923** *Simulation of a single slit using three sources*: PSSC. **924** *Pleiades*: NASA/ESA/AURA/Caltech, public domain. **924** *Radio telescope*: Wikipedia user Hajor, CC-BY-SA. **941** *Mount St. Helens*: public-domain image by Austin Post, USGS. **965** *Ozone maps*: NASA/GSFC TOMS Team. **966** *Digital camera image*: courtesy of Lyman Page. **973** *Diffacted photons*: courtesy of Lyman Page. **987** *Wicked Witch*: art by W.W. Denslow, 1900. Quote from *The Wizard of Oz*, L. Frank Baum, 1900. **998** *Werner Heisenberg*: ca. 1927, believed to be public domain. **1023** *Hindenburg*: Public domain product of the U.S. Navy.